Physics-Informed Neural Network for a Reduced WNT-HOX-Retinoic Acid Model of Colorectal Cancer

Amirkhan Aidarkhan, Nathaniel Kim

Advisor: Dr. Kataboh

Department of Mathematics & Statistics, Swarthmore College

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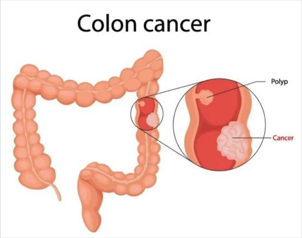

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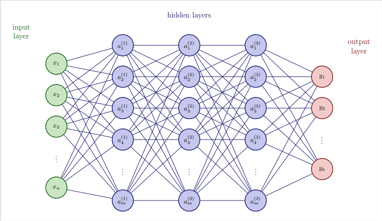


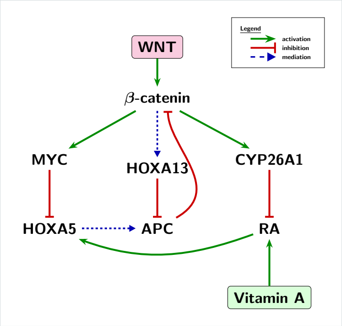
Overview

- Biology/Schematic
- Model
- Forward PINN
- Inverse PINN
- Inverse Bayesian PINN
- Next Steps

Biological Network Governing WNT–HOX–RA Signaling

Objective

Develop a reduced mathematical model describing the interactions among the WNT, HOX, and Retinoic Acid (RA) pathways during colorectal cancer progression.



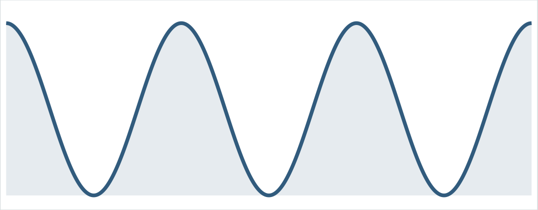
State Variables

$$\mathbf{y} = [B(t), P(t), H_5(t), H_{13}(t), M(t), R(t), C(t)]$$

B : β-catenin
P : APC
H_5 : HOXA5
H_{13} : HOXA13
M : MYC
R : Retinoic Acid
C : CYP26A1

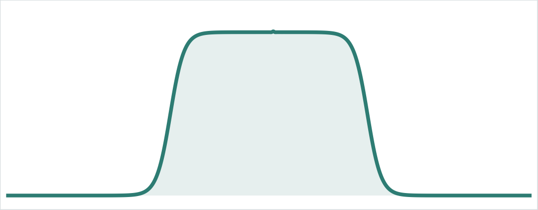
RA Input + Treatment

RA Input



$$A_d\left[1+\cos\left(\frac{2\pi t}{T_d}-\phi\right)\right]$$

ATRA Treatment



$$u_R(t)=\frac{D}{2}[\tanh q(t-t_1)-\tanh q(t-t_2)]$$

Dimensional Model

$$\frac{dB}{dt} = a_B W + v_{13}\frac{H_{13}^{n_H}}{K_{13}^{n_H} + H_{13}^{n_H}} - d_B B - k_{PB} PB - v_5 \frac{H_5 B}{K_5 + B}$$

$$\frac{dP}{dt} = a_P \frac{1 + \rho_5 H_5}{1 + \rho_B B + \rho_{13} H_{13}} - \Big(d_{P0} + d_{P1}(1 - \Theta_P)\Big) P$$

$$\frac{dH_5}{dt} = a_5 + v_R \frac{R}{K_R + R} - d_5 H_5 - v_M \frac{M H_5}{K_M + M}$$

$$\frac{dH_{13}}{dt} = a_{13} + v_{B13} \frac{B^{n_B}}{K_{B13}^{n_B} + B^{n_B}} + v_{M13} \frac{M^{n_M}}{K_{M13}^{n_M} + M^{n_M}} - d_{13} H_{13}$$

$$\frac{dM}{dt} = a_M + v_{BM} \frac{B^{n_B}}{K_{BM}^{n_B} + B^{n_B}} - d_M M$$

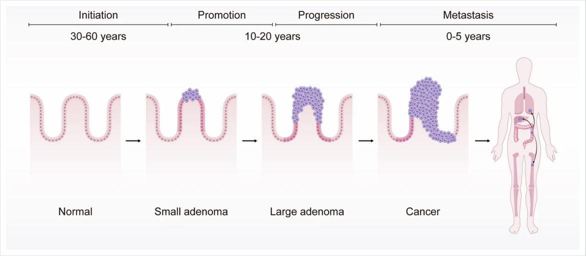
$$\frac{dR}{dt} = u_0 + A_d \left[1 + \cos\left(\frac{2\pi t}{T_d} - \phi\right)\right] + \frac{D}{2}\Big[\tanh(q(t - t_1)) - \tanh(q(t - t_2))\Big] - d_R R - k_{CR} C R$$

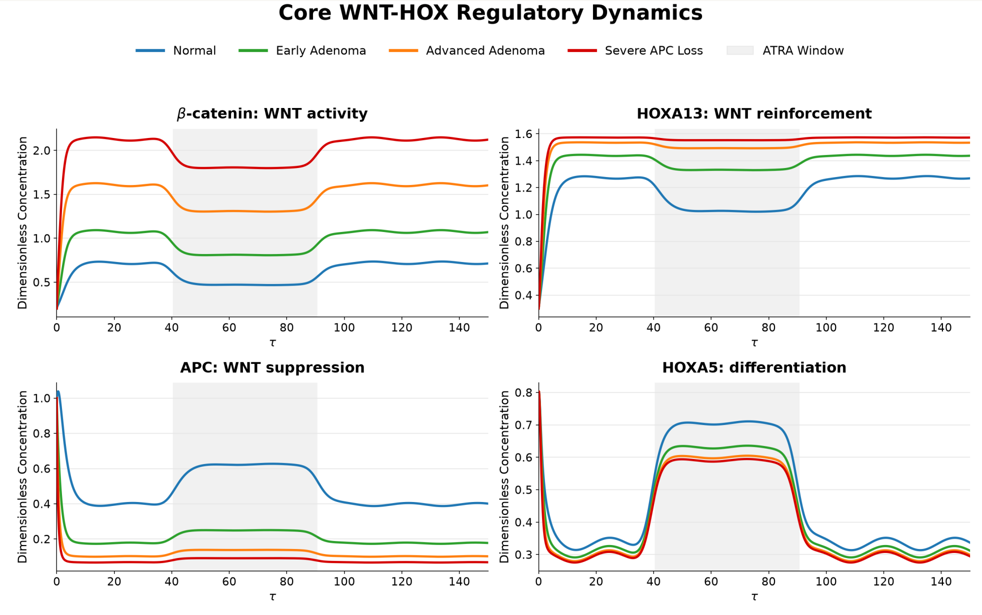
$$\frac{dC}{dt} = a_C + v_{RC} \frac{R}{K_{RC} + R} + v_{BC} \frac{B^{n_B}}{K_{BC}^{n_B} + B^{n_B}} - d_C C$$

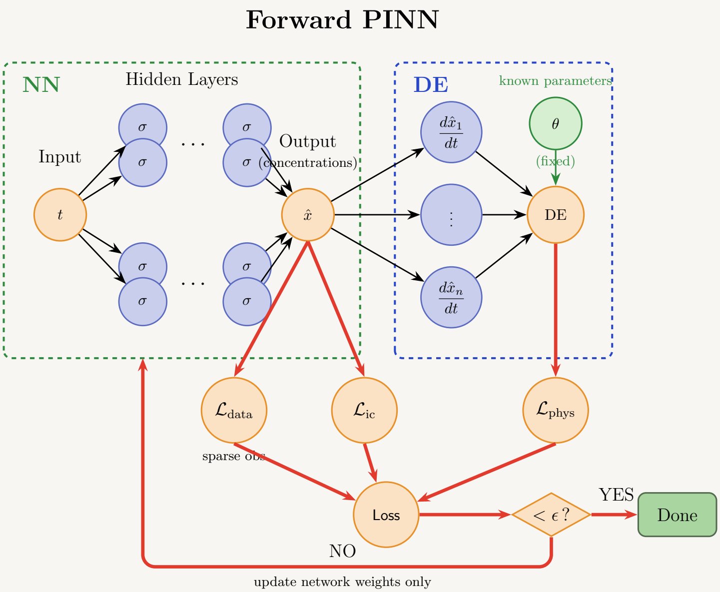
4 Regimes

Disease progression is modeled by increasing WNT signaling (W) and progressive loss of APC function (θ_P), mimicking CRC progression.

- "Normal"
  - $W = 0.8$, $\theta_P = 1.0$
- "Early Adenoma"
  - $W = 1.0$, $\theta_P = 0.75$
- "Advanced Adenoma"
  - $W = 1.5$, $\theta_P = 0.5$
- "Severe APC Loss"
  - $W = 2.0$, $\theta_P = 0.25$

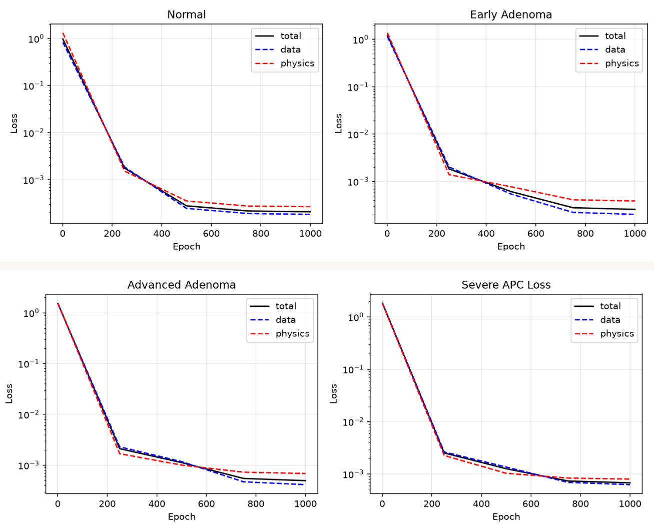


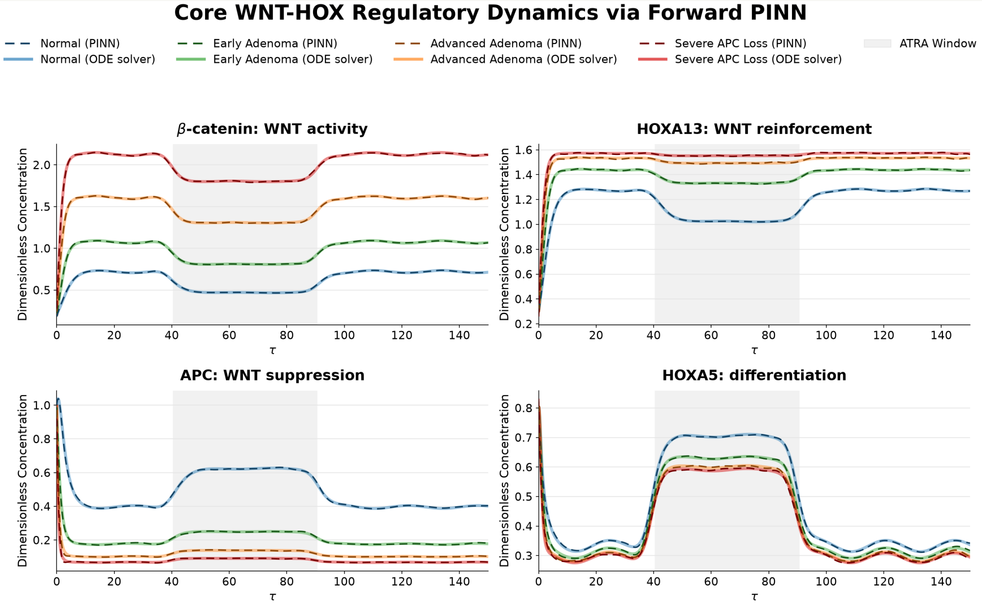

Core WNT-HOX Regulatory Dynamics
Normal
Early Adenoma
Advanced Adenoma
Severe APC Loss
ATRA Window
β-catenin: WNT activity
HOXA13: WNT reinforcement
APC: WNT suppression
HOXA5: differentiation
Dimensionless Concentration
τ



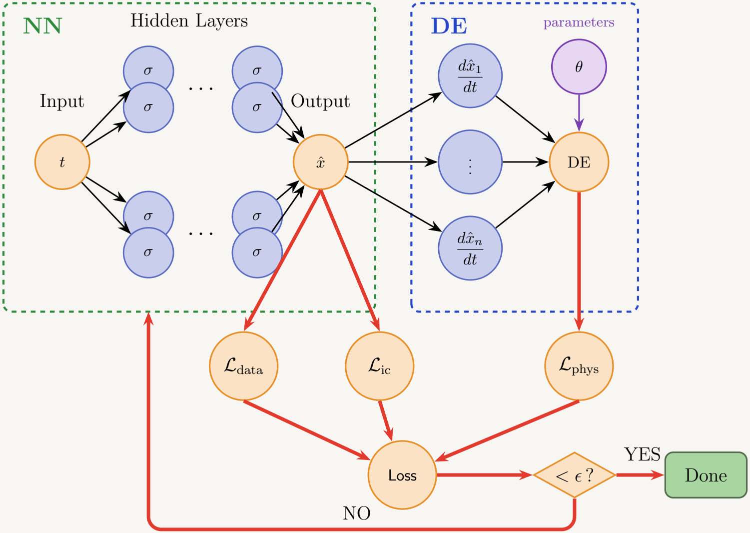
Forward PINN
NN
Hidden Layers
Input
Output
(concentrations)
t
σ
$\cdots$
$\hat{x}$
DE
known parameters
θ
(fixed)
$\frac{d\hat{x}_1}{dt}$
$\vdots$
$\frac{d\hat{x}_n}{dt}$
DE
$\mathcal{L}_{\text{data}}$
sparse obs
$\mathcal{L}_{\text{ic}}$
$\mathcal{L}_{\text{phys}}$
Loss
$< \epsilon\,?$
YES
Done
NO
update network weights only

Forward PINN Loss

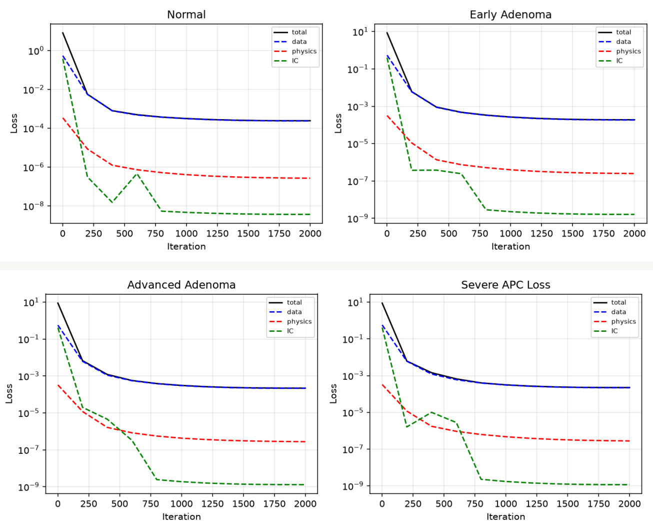


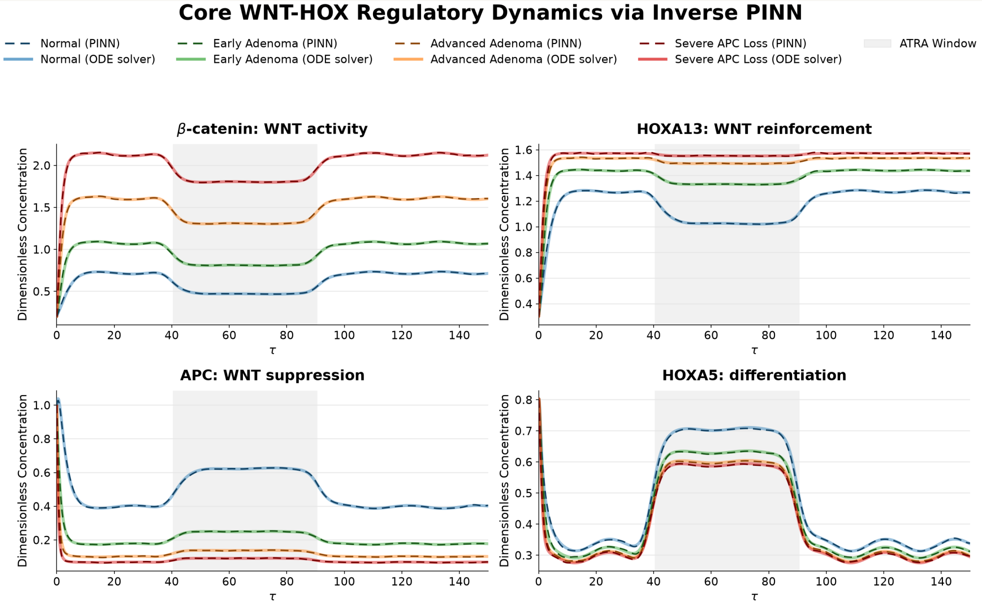

Core WNT-HOX Regulatory Dynamics via Forward PINN
Normal (PINN)
Normal (ODE solver)
Early Adenoma (PINN)
Early Adenoma (ODE solver)
Advanced Adenoma (PINN)
Advanced Adenoma (ODE solver)
Severe APC Loss (PINN)
Severe APC Loss (ODE solver)
ATRA Window
β-catenin: WNT activity
HOXA13: WNT reinforcement
APC: WNT suppression
HOXA5: differentiation
Dimensionless Concentration
τ

Inverse PINN



Inverse PINN Loss





Core WNT-HOX Regulatory Dynamics via Inverse PINN
Normal (PINN)
Normal (ODE solver)
Early Adenoma (PINN)
Early Adenoma (ODE solver)
Advanced Adenoma (PINN)
Advanced Adenoma (ODE solver)
Severe APC Loss (PINN)
Severe APC Loss (ODE solver)
ATRA Window
β-catenin: WNT activity
HOXA13: WNT reinforcement
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Dimensionless Concentration
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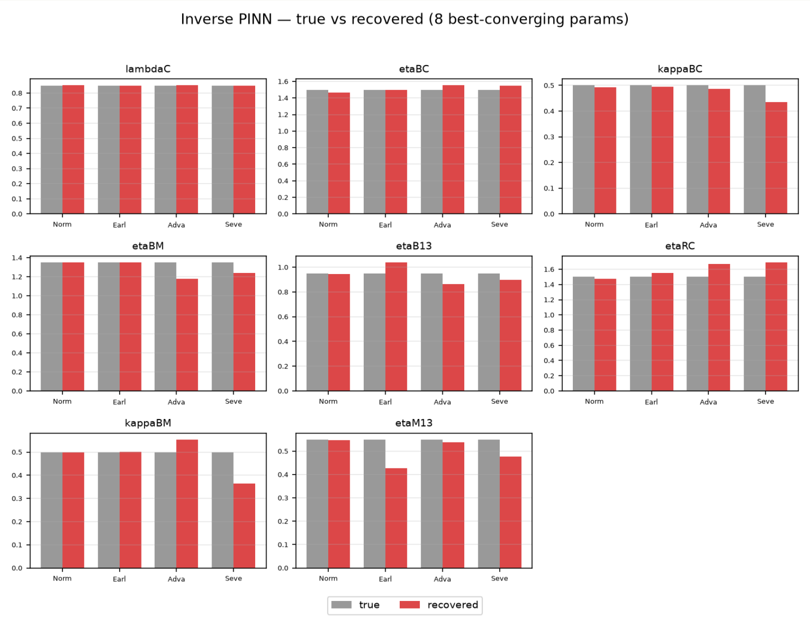
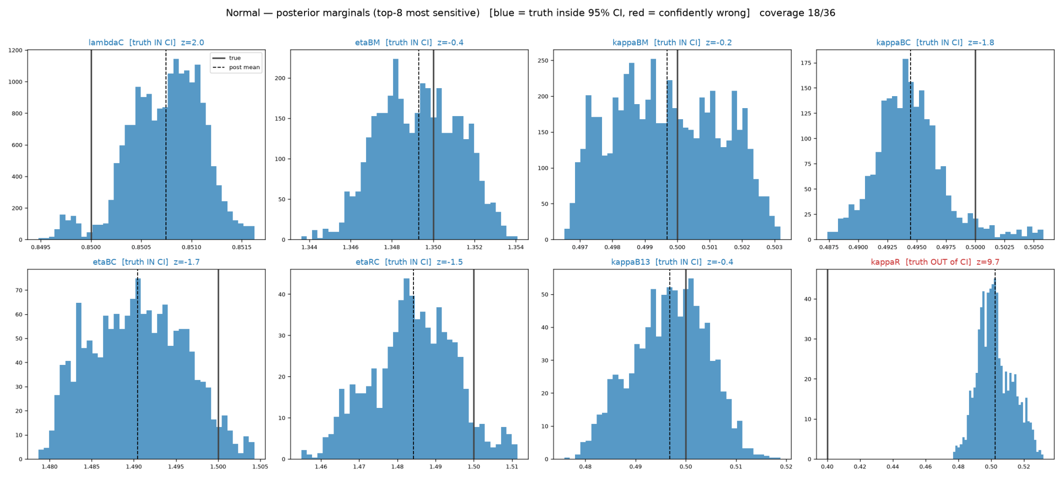
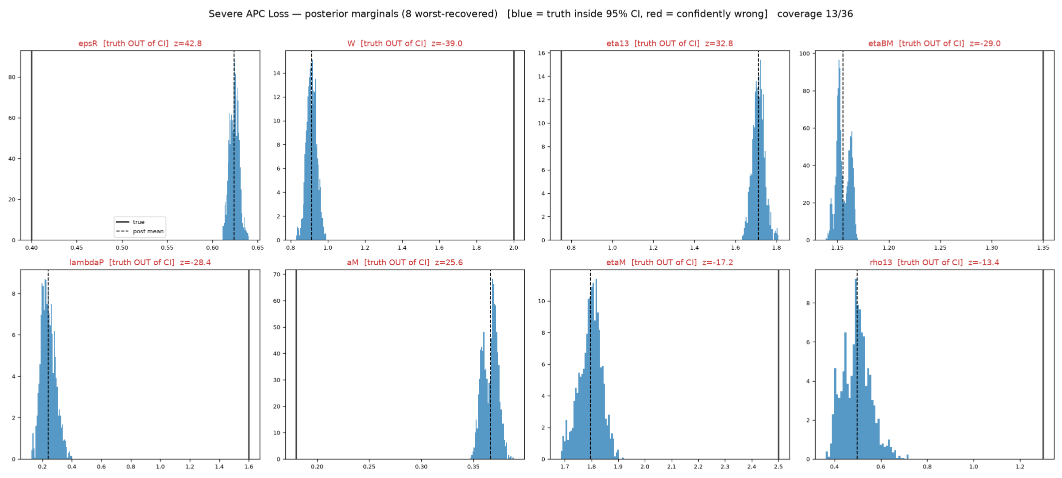

Inverse PINN — true vs recovered (8 best-converging params)
lambdaC
etaBC
kappaBC
etaBM
etaB13
etaRC
kappaBM
etaM13
Norm
Earl
Adva
Seve
true
recovered

Inverse Bayesian PINN - uncertainty quantification



Inverse Bayesian PINN - continued



Parameter Recovery Comparison

	Bayesian inverse PINN	Inverse PINN	ODE-fit
“Normal”	**19**	17	18/36
“Early Adenoma”	**20**	16	17/36
“Advanced Adenoma”	11	10	**13/36**
“Severe APC Loss”	8	7	**11/36**

Next Steps

- Improve Bayesian PINN
  - Investigate limitations
- Hybrid Mechanistic-Neural Model
- Stochastic Modeling

Key References

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6. **Tancik *et al.* (2020).** Fourier features let networks learn high frequency functions in low dimensional domains. *NeurIPS* **33**, 7537–7547.
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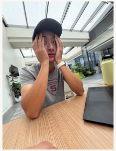
Thank you!

Questions? (we will try to answer) 😅

PK calling Amir + Nate to check in on our progress…

4th diet coke of the day + Celsius b/c his sleep schedule is so cooked



Don't know what in the world is going on b/c he spent more hours watching the world cup than working

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